MBI806B BUSINESS DATA ANALYTICS WITH AI AND ML

Dashboard Design Studio

A group activity: design a one-page dashboard for a client, in a free tool

Lecturer **Yasas Sri Wickramasinghe** Time **About 90 minutes**

Teams **Five, with four to six people each**

You need **A laptop, a free Google account, pencil and paper**

What you are doing

A dashboard is a decision on one screen. Each team gets a different client, a different reader and a different screen. You decide what goes on the page, which chart answers each question and where each one sits, then mock it up. You are not building it yet. You are designing it.

By the end you should be able to

- state who a dashboard is for and the decision it supports
- turn a decision into three to five questions
- choose the right chart for each question, and say what you rejected
- lay out a one-page dashboard so the eye lands on the right thing first
- use colour and titles to direct attention
- test a design on someone else in five seconds.

Contents

Part 1	The tool: a free, easy mockup tool
Part 2	How to design one: five steps, the chart cheat sheet, six habits
Part 3	Layout skeletons, the session plan and what you hand in
Part 4	Feedback: the five-second test and a checklist
Team 1	Kōtuku Café Group: The Monday-morning sales review
Team 2	Harbourview Medical Centre: The missed-appointments dashboard
Team 3	Tāmaki Courier Co.: The control-room wall screen
Team 4	Rimu Outdoor Gear: Where do our customers come from, and where do we lose them?
Team 5	Tūī Valley Council: The public recycling dashboard

For the lecturer

Each team's section is four pages: the brief, the data, a sketch page and a rationale table. Print the shared pages once, then print only each team's four pages and hand one set to each team. The figures in the briefs add up across tables, so two teams checking the same number will agree. The model answers are a separate PDF: hold it back until the teams have presented.

© Yasas Sri Wickramasinghe

PART 1

Use a free, easy mockup tool

Pick the tool your whole team can already open in the next minute. The tool is not the point.

OUR PICK

Google Slides

Free with any Google account, nothing to install, and the whole team edits the same file at once. A slide is a ready-made 16:9 canvas, and Insert → Chart gives you real bar, line and pie charts you can edit with your own numbers. Everybody already knows how to use it.

Watch for: It has no dashboard templates, so you build the layout from rectangles and text boxes. That is a feature for this activity: you decide the layout instead of filling in a template.

`slides.google.com`

PRETTIER, IF SOMEONE KNOWS IT

Canva

Free plan, runs in the browser, lots of dashboard-style templates, and a built-in chart tool where you type the numbers in.

Watch for: Templates tempt teams to fill boxes instead of choosing charts. Check that your team can edit the same design at once on the free plan before you start, and note that some elements are paid.

`www.canva.com`

MOST CONTROL, STEEPEST TO LEARN

Figma

The free plan is what professional designers use for screen mockups, with precise grids and alignment.

Watch for: There is a learning curve, and the free plan limits how many files a team can share, so only pick it if a team member already uses it.

`www.figma.com`

FREE, NO SIGN-UP

Excalidraw (for the sketch)

A shared whiteboard that looks hand-drawn. Perfect for the five-minute layout sketch before you open a mockup tool. Paper and pencil work just as well.

Watch for: It cannot draw charts from data. Use it for the sketch, then move to Slides, Canva or Figma.

`excalidraw.com`

Free plans and limits change. Check the current limits on the tool's own site before you commit your team to it.

Set up a dashboard canvas in Google Slides (about 5 minutes)

1. Go to slides.google.com and create a blank presentation. Name it with your team number and client, for example "Team 3 · Tāmaki Courier".
2. Check the slide size under File → Page setup. Choose the size that matches your brief: widescreen 16:9 for a desktop, TV or projector, or a custom portrait size for a phone.
3. Turn on guides and snapping from the View menu, so rectangles and charts line up on a grid.
4. Draw one rectangle per visual first, in light grey, and label each with the question it answers. This is your layout. Move things around until the priority order is right.
5. Replace each grey rectangle with a real chart: Insert → Chart, then choose bar, column, line or pie. Use "Open source" to edit the numbers in the linked Google Sheet and paste in your data table.
6. Format for the reader: one strong colour plus greys, big enough text for the screen, direct labels instead of legends, and a title that states the finding.
7. Share the file with your teammates as editors so you can all work at once. Keep a final copy as a PDF or PNG for the presentation.

Why not start in Power BI?

Power BI will make a chart for you in ten seconds, and the chart it picks is rarely the right one. Designing on a blank slide first makes you decide before the software decides for you. When the design is settled, building it is the easy part.

© Yasas Sri Wickramasinghe

PART 2

How to design one: five steps, in this order

1. **Name the reader and the decision.** Write one sentence: "[Who] needs to decide [what], looking at [which screen], [how often]." If you cannot write it, you cannot design the dashboard. Every later choice comes back to this sentence.
2. **Write the questions.** List three to five questions the reader must be able to answer within five seconds. Each question will become one visual. A visual that answers no question is decoration, and it goes.
3. **Match each question to a chart.** Use the cheat sheet below. Decide the chart from the type of question, not from what looks interesting. Write down the chart you rejected and why. That sentence is half of your presentation.
4. **Sketch the layout on paper.** Five minutes, pencil, no computer. Put the most important thing top-left, group visuals that answer related questions, and keep the whole screen to six to nine items. Decide the layout before choosing colours.
5. **Mock it up, then test it.** Build the sketch in your tool using the real numbers from your data tables. Then run the five-second test: show another team the screen for five seconds, hide it, and ask what the main message was.

Chart cheat sheet: pick by the type of question

The question is about	Reach for	Be careful with
Compare categories (cafés, channels, depots)	Horizontal bar chart, sorted from largest to smallest	Pie or donut with more than four slices, 3D charts, bright rainbow colours
Show a trend over time	Line chart, with a target or last-year line where one exists	A y-axis that starts well above zero on a bar, columns for 60 data points
Compare a result with a target	A bar with a target marker, or a progress bar, with the number written next to it	Speedometer gauges, red/green traffic lights on their own
Show part of a whole (two to four parts)	A single stacked bar, or a donut with direct labels	Pies with many thin slices, two pies side by side to compare
Find a pattern across two categories (day × time)	Heatmap with the values printed in the cells	Twelve line charts, a rainbow colour scale
Show where people drop out of a process	A funnel, or horizontal bars with the percentage kept at each step	A table of five numbers with no visual
Show one important number	A KPI tile: the number, the unit, and a comparison (target, last period)	A bare number that nobody can judge
Relate two measures (spend and revenue)	Scatter plot, labelled points	A dual-axis line chart, where the crossing point means nothing

Six habits of a dashboard that gets read

Top-left is read first

Put the single most important item there. Readers scan roughly in a Z or an F shape, so size and position carry the priority.

Six to nine visuals, not twenty

Every extra chart steals attention from the others. If a team cannot say which question a visual answers, remove it.

Say the finding in the title

"Saturday revenue is double Monday's" tells the reader what to see. "Revenue by day" makes them work it out.

Use colour to point, not to decorate

Grey for everything, one strong colour for the thing that matters. Add a second cue (icon, label, position) so colour is never the only signal.

Compare with something

A number without a target, a last period or a benchmark cannot be judged. Show the comparison next to it.

Align, space, and remove clutter

Snap everything to a grid, leave white space between groups, and delete gridlines, borders and 3D effects that carry no information.

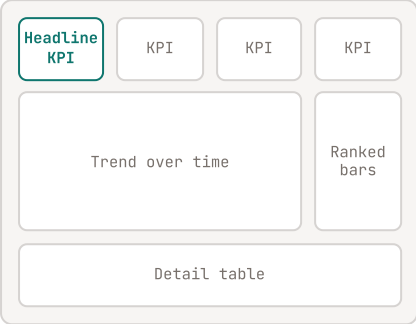
© Yasas Sri Wickramasinghe

PART 3

Layout, session plan and hand-in

Three skeletons to start from

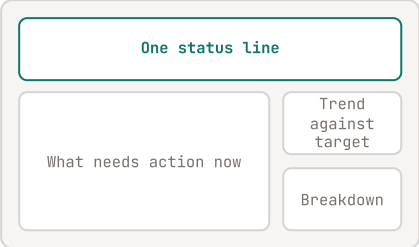
The outlined box is where the eye lands first. Whatever you put there is the most important thing on the page.



Summary board

For: A manager reviewing last week or last month

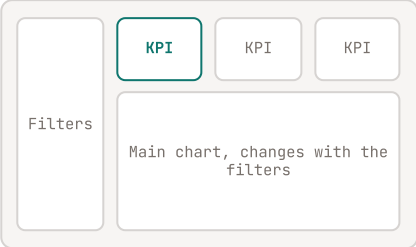
Numbers first, then the trend that explains them, then the detail for anyone who wants it.



Glance board

For: A team looking up from other work, from a distance

Almost nothing on it. The status says whether to act, and the biggest area says where.



Explorer

For: An analyst who will filter and click

Controls on the left, results on the right. Only suits a reader who has time and a reason to explore.

The session, 90 minutes

Time	Step	What to do
10 min	**Form teams and read your brief**	Five teams of four to six. Each team takes the next scenario, reads the brief out loud, and agrees the one-sentence reader and decision statement.
10 min	**Critique the dashboard they have today**	Go through the "what the current dashboard gets wrong" list. For each item, say why it fails the reader. This is your design checklist.
15 min	**Questions, then charts**	Write the questions, then choose a chart for each from the cheat sheet. Write down one chart you rejected.
10 min	**Sketch the layout on paper**	Pencil on the sketch page. Priority top-left, related visuals together. Photograph it.
25 min	**Build the mockup**	Divide the work: one person per two or three visuals, one person owns layout and consistency. Use the real numbers from your data tables.
5 min	**Five-second test**	Swap with another team. Show for five seconds, hide, and ask what the main message was. Write down what they said.
15 min	**Present and give feedback**	Three minutes per team: the reader, the layout, and two design decisions with the chart you rejected. Then two minutes of feedback each.

What each team hands in

A one-page mockup	The dashboard as the reader will see it, built from the real numbers in your data tables, in the tool you chose.
A design rationale table	One row per visual: the question it answers, the chart you chose, the chart you rejected and why, and how you used colour.
A three-minute pitch	Who the reader is, how the layout leads their eye, and two decisions you are proud of. Everyone in the team speaks.

© Yasas Sri Wickramasinghe

PART 4

Give feedback that changes the design

You are not marking each other. You are finding what a real reader would trip over.

The five-second test

Show another team's dashboard for exactly five seconds, then hide it. Ask: "What was it about? What should the reader do?" If they cannot say, the layout is not leading the eye. Write down what they said, word for word, before you explain anything.

Three sticky notes

When you walk around the other teams' screens, leave three notes on each: **I noticed** (where my eye went first), **I got lost at** (what slowed me down), and **One change** (the single edit that would help most).

Checklist for any dashboard

- [ ] Five-second test: could a stranger say the main message after five seconds?
- [ ] Is the most important item in the most prominent position?
- [ ] Can you point to the question each visual answers?
- [ ] Is every chart the right type for its question?
- [ ] Does every number have something to compare against?
- [ ] Do the titles state findings rather than topics?
- [ ] Is colour used with purpose, with a second cue for status?
- [ ] Could the screen be read on the device and from the distance in the brief?
- [ ] Has anything been included that the reader does not need?
- [ ] Is anything shown that should not be, such as identifying information?

Notes from the five-second test

#	Team tested	What they said it was about	What was missed
1			
2			
3			
4			

Next step: when your design is settled, build it for real in Power BI.

© Yasas Sri Wickramasinghe

TEAM 1 Kōtuku Café Group · Brief

The Monday-morning sales review

Six cafés across Auckland · hospitality

READER	SCREEN	HOW OFTEN
Hana Rātana, the group's regional manager	A tablet, held in her hand in the weekly meeting with café managers	Every Monday, for ten minutes

Kōtuku runs six cafés. Every Monday Hana meets the six café managers to review last week and decide where to move people and stock. Today she gets a spreadsheet with 40 columns, printed on three sheets of paper.

She wants one screen she can hold up in the meeting. The managers should be able to see in seconds which café is ahead of target, which is behind, and whether the group as a whole is heading in the right direction.

THE DECISION THIS DASHBOARD SUPPORTS

Which cafés need extra staff or stock this week, and which are on track?

The reader must be able to answer, in five seconds

1. How is each café doing against its quarterly target?
2. Is group revenue going up or down over the quarter, and when did it change?
3. Which days of the week are busiest, so rosters can match demand?
4. Which café has the weakest average spend per customer?

What their dashboard gets wrong today

- A 3D pie chart splits group revenue between the six cafés. The slices are nearly the same size and cannot be compared.
- Every café is shown in a different bright colour, so colour carries no meaning.
- Red and green traffic lights show "behind" and "ahead", which a colour-blind manager cannot tell apart.
- The headline number is $1,728,800 with no target, no last-quarter figure and no arrow, so nobody knows whether it is good.

Your design must

- Landscape tablet, about 1024 × 768. Nothing smaller than a readable size.
- No more than six visuals. Hana has ten minutes.
- Must work for a colour-blind reader: never use colour as the only signal.

STRETCH CHALLENGE

The managers keep asking "is this a bad week or a bad café?" Add one element that separates a café problem from a group-wide dip, without adding a seventh chart.

© Yasas Sri Wickramasinghe

Your data

Copy these numbers into your chart tool. Work out any rates and totals you need from the counts. All figures are invented for this activity.

Table A · This quarter, by café

Average spend per customer is revenue divided by customers.

Café	Revenue	Target	Customers	Avg spend
Ponsonby	$412,000	$400,000	61,200	$6.73
Newmarket	$368,500	$390,000	58,900	$6.26
Takapuna	$301,200	$300,000	40,100	$7.51
Mt Eden	$254,800	$280,000	38,300	$6.65
Botany	$223,400	$250,000	36,700	$6.09
Henderson	$168,900	$210,000	29,800	$5.67

Table B · Group revenue by week (13-week quarter)

Weeks 11 and 12 were school holidays.

Week	Revenue	Week	Revenue
Week 1	$128,000	Week 8	$139,400
Week 2	$130,500	Week 9	$137,100
Week 3	$127,900	Week 10	$141,300
Week 4	$133,200	Week 11	$129,500
Week 5	$135,600	Week 12	$124,600
Week 6	$132,800	Week 13	$132,000
Week 7	$136,900		

Table C · Average group revenue per day of the week

Rounded to the nearest $100.

Day	Revenue	Day	Revenue
Monday	$13,600	Friday	$20,300
Tuesday	$14,400	Saturday	$28,200
Wednesday	$15,100	Sunday	$24,900
Thursday	$16,500		

© Yasas Sri Wickramasinghe

Team 1 · Kōtuku Café Group: sketch your layout

Pencil, five minutes, then photograph it

Tablet, landscape (4:3)

Reader and decision, in one sentence

The most important item (top-left)

Notes: what moved, and why

© Yasas Sri Wickramasinghe

Design rationale

One row for every visual on your dashboard. This table is half of your three-minute pitch: it shows that each chart was chosen, not defaulted to.

READER AND DECISION

"Hana Rātana needs to decide … looking at … , …"

#	Question it answers	Chart chosen	Chart rejected, and why	How colour is used
Ex.	Which classes at a gym are over-full or empty?	Horizontal bars of attendance as a share of capacity, sorted, with a 100% line	Pie chart: twelve thin slices, and a slice cannot show capacity	Grey bars. Accent only above 90% full
1				
2				
3				
4				
5				
6				
7				
8				

© Yasas Sri Wickramasinghe

TEAM 2 Harbourview Medical Centre · Brief

The missed-appointments dashboard

A GP practice with eight clinicians · health

READER	SCREEN	HOW OFTEN
Maria Fonoti, the practice manager	A desktop monitor in her office, shared with the clinical lead on screen	The first working day of each month, for a 30-minute review

When a patient does not turn up, the practice calls it a DNA (did not attend). Over the last six months Harbourview had 5,780 bookings and 456 DNAs. Each one wastes a clinician's time and delays someone else who needed the slot.

The practice is deciding where to spend a small budget: text reminders, a different booking system, or overbooking the worst slots. Maria needs a dashboard that shows where and when DNAs happen, in a way the clinical lead trusts.

THE DECISION THIS DASHBOARD SUPPORTS

Where is the DNA problem biggest, and which fix should the practice spend its budget on?

The reader must be able to answer, in five seconds

1. What is the overall DNA rate, and is it getting better or worse month to month?
2. Which days and times have the highest DNA rate?
3. Which appointment types are missed most often?
4. Do text reminders make a difference?

What their dashboard gets wrong today

- A table lists each repeat no-show by patient name and NHI number. Anyone looking at the shared screen can read it.
- The chart shows raw DNA counts, so the busiest months and slots always look worst, even when their rate is no higher.
- Fourteen months of data sit on one line chart with a y-axis that starts at 40, which makes small changes look dramatic.
- Friday afternoon looks like the worst slot, but it is based on a handful of bookings. Nothing on the screen says so.

Your design must

- Desktop monitor, 1920 × 1080, shared with other people in the room.
- No patient names, IDs or anything that could identify a person. Show groups only.
- Rates and counts must both be available. A rate without its size can mislead.

STRETCH CHALLENGE

The clinical lead asks whether the afternoon difference is real or just chance. Show uncertainty or sample size so a small base is not read as a finding.

© Yasas Sri Wickramasinghe

Your data

Copy these numbers into your chart tool. Work out any rates and totals you need from the counts. All figures are invented for this activity.

Table A · By month

Month	Booked	DNAs
April	940	70
May	960	74
June	1,010	85
July	1,020	90
August	950	75
September	900	62

Table B · By day and time of day (April to September)

Each cell is DNAs / booked. Friday afternoon is a short session because the practice closes at 3pm.

Day	Early (8–10)	Midday (10–2)	Afternoon (2–5)
Monday	22 / 310	38 / 520	41 / 410
Tuesday	20 / 290	35 / 540	37 / 430
Wednesday	17 / 280	31 / 500	39 / 390
Thursday	21 / 300	37 / 530	40 / 420
Friday	19 / 260	33 / 450	26 / 150

Table C · By appointment type

Type	Booked	DNAs
Routine check	2,310	139
Prescription review	1,090	49
Child immunisation	760	29
Mental health and wellbeing	640	80
Allied health referral	980	159

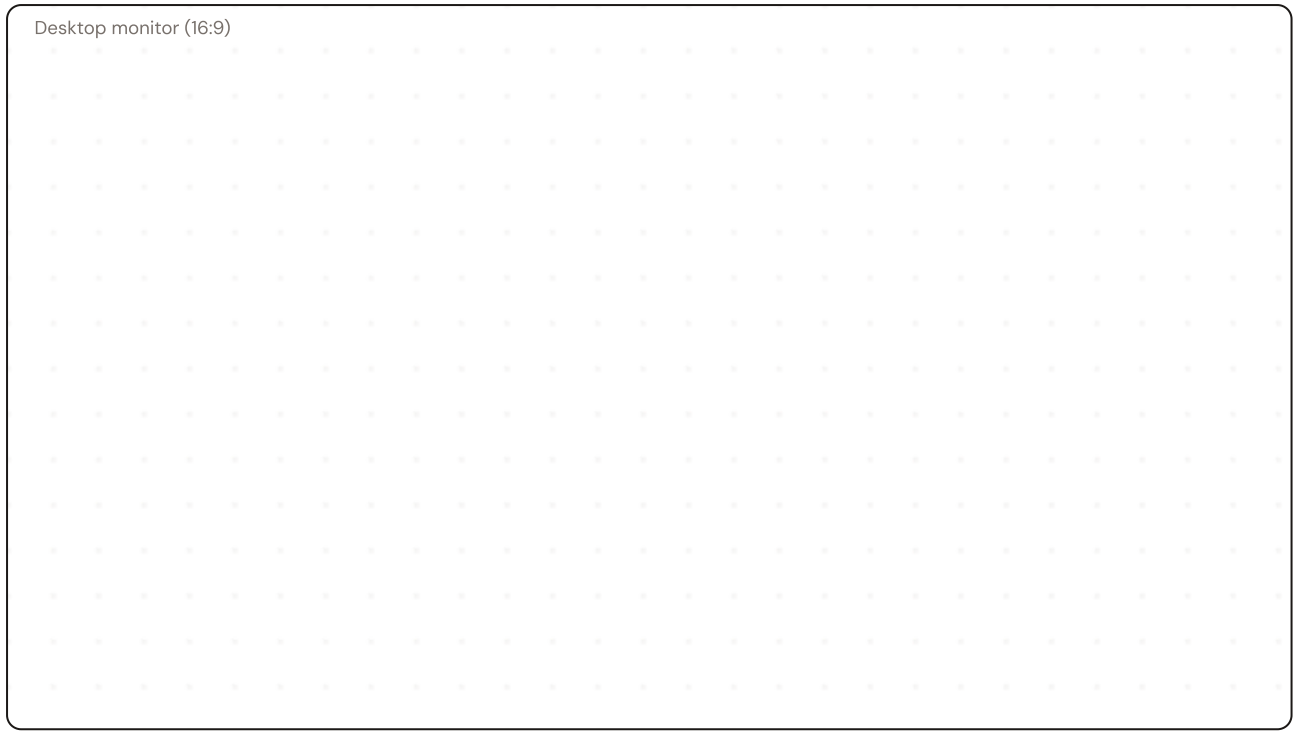
Table D · Reminders

Group	Booked	DNAs
Text reminder sent	3,900	218
No reminder (no mobile on file)	1,880	238

© Yasas Sri Wickramasinghe

Team 2 · Harbourview Medical Centre: sketch your layout

Pencil, five minutes, then photograph it



Reader and decision, in one sentence

The most important item (top-left)

Notes: what moved, and why

© Yasas Sri Wickramasinghe

Design rationale

One row for every visual on your dashboard. This table is half of your three-minute pitch: it shows that each chart was chosen, not defaulted to.

READER AND DECISION

"Maria Fonoti needs to decide … looking at … , …"

#	Question it answers	Chart chosen	Chart rejected, and why	How colour is used
Ex.	Which classes at a gym are over-full or empty?	Horizontal bars of attendance as a share of capacity, sorted, with a 100% line	Pie chart: twelve thin slices, and a slice cannot show capacity	Grey bars. Accent only above 90% full
1				
2				
3				
4				
5				
6				
7				
8				

© Yasas Sri Wickramasinghe

TEAM 3 Tāmaki Courier Co. · Brief

The control-room wall screen

Same-day parcel delivery, five depots · logistics

READER	SCREEN	HOW OFTEN
The duty controller and two dispatchers, in the control room	A 55-inch TV on the wall, read from three to five metres away	All day. Glanced at many times an hour, refreshed every few minutes

It is 2pm. Tāmaki Courier has 7,600 parcels to deliver today across five depots and 5,350 are delivered. The company promises 95% of parcels on time.

The controllers do not read this screen, they glance at it. They need to know in two seconds whether anything needs action right now, and if so where. Everything else is secondary.

THE DECISION THIS DASHBOARD SUPPORTS

Which depot or van needs a controller's attention right now, and is the day going to hit the 95% promise?

The reader must be able to answer, in five seconds

1. Is the company on track for 95% on time today?
2. Which depot is in trouble, and how bad is it?
3. Are deliveries keeping pace through the day?
4. Why are delivery attempts failing, and is that getting worse?

What their dashboard gets wrong today

- Forty-two small charts fill the screen. From the back of the room none of the labels can be read.
- Everything is the same size and the same blue, so a van that is two hours late looks like a van that is two minutes late.
- The company-wide average hides the fact that one depot is far worse than the rest.
- There is no target line anywhere, so "92.5%" means nothing without knowing it should be 95%.

Your design must

- Wall-mounted TV, 1920 × 1080, read from several metres. Text must still be legible at a distance.
- Designed to be glanced at: the key message must be visible in two seconds, with no hovering or clicking.
- Colour may signal status, but it needs a second cue (a symbol, a label or a position) for colour-blind staff.

STRETCH CHALLENGE

Design what the screen looks like when everything is fine. A dashboard that shouts all day gets ignored. How does your layout stay quiet until something is wrong?

© Yasas Sri Wickramasinghe

Your data

Copy these numbers into your chart tool. Work out any rates and totals you need from the counts. All figures are invented for this activity.

Table A · Depots, 2pm today

On-time % is delivered on time divided by delivered so far. The promise is 95%.

Depot	Parcels today	Delivered	On time	On-time %	Failed	Vans out	Vans 30+ min late
East Tāmaki	2,400	1,680	1,596	95.0%	38	22	2
Auckland Central	1,850	1,420	1,292	91.0%	41	18	4
Albany	1,300	910	882	96.9%	14	12	0
Manukau	1,150	700	560	80.0%	36	11	5
Henderson	900	640	621	97.0%	9	8	1

Table B · Parcels delivered per hour today (all depots)

Hour	Delivered
8–9am	430
9–10am	850
10–11am	1,010
11am–12pm	1,090
12–1pm	1,060
1–2pm	910

Table C · Company on-time % for the last 13 days (today is the last)

Day	On-time %	Day	On-time %
Day 1	94.1%	Day 8	96.1%
Day 2	95.3%	Day 9	95.4%
Day 3	95.8%	Day 10	94.9%
Day 4	96.0%	Day 11	95.7%
Day 5	94.7%	Day 12	96.2%
Day 6	93.2%	Today	92.5%
Day 7	95.5%		

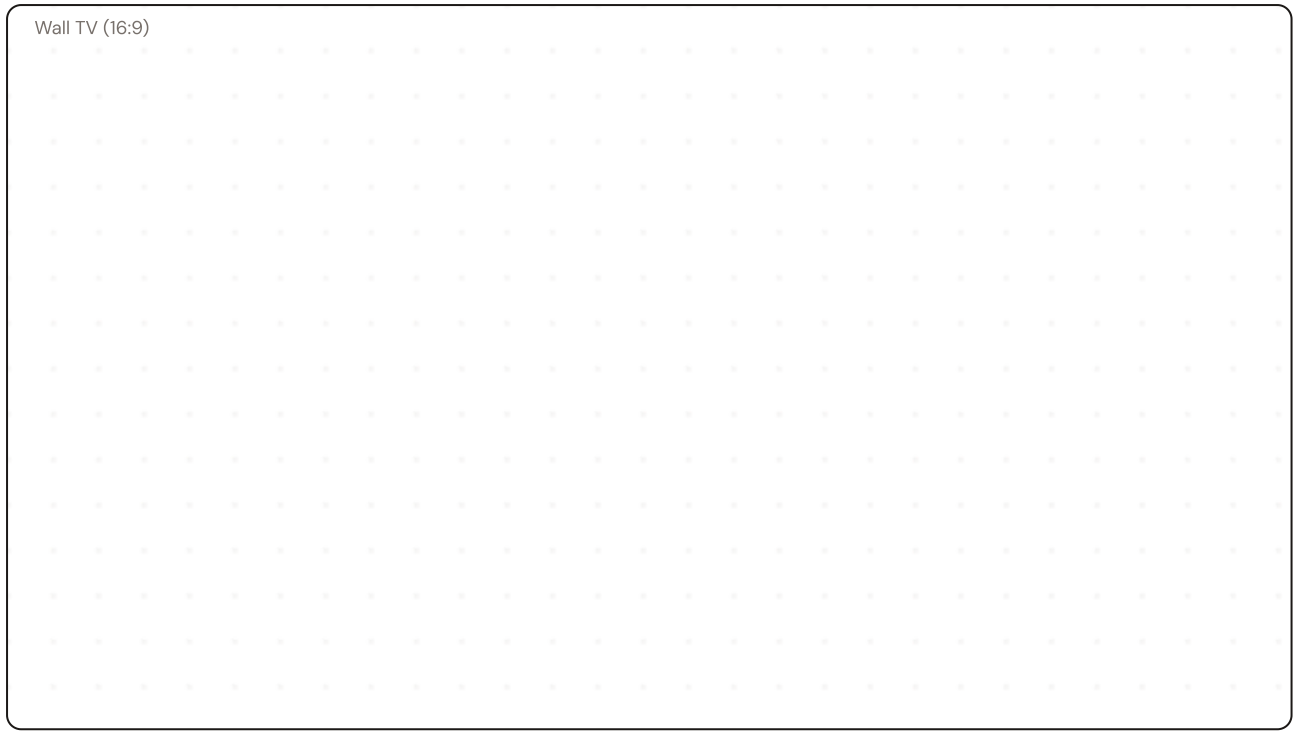
Table D · Why delivery attempts failed today

Reason	Failed attempts
Nobody home	61
Address problem	34
Access or gate problem	22
Customer refused	12
Damaged or other	9

© Yasas Sri Wickramasinghe

Team 3 · Tāmaki Courier Co.: sketch your layout

Pencil, five minutes, then photograph it



Reader and decision, in one sentence

The most important item (top-left)

Notes: what moved, and why

© Yasas Sri Wickramasinghe

Design rationale

One row for every visual on your dashboard. This table is half of your three-minute pitch: it shows that each chart was chosen, not defaulted to.

READER AND DECISION

"The duty controller and two dispatchers needs to decide … looking at … , …"

#	Question it answers	Chart chosen	Chart rejected, and why	How colour is used
Ex.	Which classes at a gym are over-full or empty?	Horizontal bars of attendance as a share of capacity, sorted, with a 100% line	Pie chart: twelve thin slices, and a slice cannot show capacity	Grey bars. Accent only above 90% full
1				
2				
3				
4				
5				
6				
7				
8				

© Yasas Sri Wickramasinghe

TEAM 4 Rimu Outdoor Gear · Brief

Where do our customers come from, and where do we lose them?

An online shop selling tramping and snow gear · e-commerce

READER	SCREEN	HOW OFTEN
Tom Nguyen, the marketing manager, presenting to the CEO	A laptop, then projected onto a meeting-room screen	Monthly, as the first ten minutes of the budget meeting

Last month (September) Rimu had 184,000 visits, 6,900 orders and $839,030 of revenue. The CEO wants to move next quarter's marketing budget, but every channel reports its own flattering numbers.

Tom's dashboard has to show the whole journey from a visit to a purchase, compare the channels fairly, and show whether the business is seasonal, so the budget goes where it earns the most.

THE DECISION THIS DASHBOARD SUPPORTS

Which marketing channels get more budget next quarter, and where in the buying journey are we losing the most customers?

The reader must be able to answer, in five seconds

1. How many visitors become buyers, and where do most drop out?
2. Which channels earn the most revenue for the money spent, not just the most visits?
3. How does this month compare with the same month and the months around it?
4. What is the single number the CEO should remember from this page?

What their dashboard gets wrong today

- The page opens with total page views, followers and subscribers. All three are big and growing, and none of them says anything about revenue.
- Channel revenue is shown as a pie. Paid social looks like a reasonable slice, so no one notices it costs $22,500 and returns $54,600.
- The journey is a table of five numbers. The 72% who add to cart and do not buy is never highlighted.
- Two y-axes on the revenue chart (revenue and orders) make the lines cross at a meaningless point.

Your design must

- Laptop and projector, 16:9. Must still be readable when projected in a bright room.
- Exactly one headline number. A CEO will remember one thing.
- Every chart needs a title that states the finding, not just the topic.

STRETCH CHALLENGE

Tom says "the CEO will ask what happens if we move $10,000 from paid social to email." Show what you would need to add, or what you must warn the CEO not to assume.

© Yasas Sri Wickramasinghe

Your data

Copy these numbers into your chart tool. Work out any rates and totals you need from the counts. All figures are invented for this activity.

Table A · The buying journey, September

Step	People	% of visits
Visited the site (sessions)	184,000	100.00%
Viewed a product	121,400	65.98%
Added to cart	24,800	13.48%
Started checkout	12,100	6.58%
Completed a purchase	6,900	3.75%

Table B · By channel, September

Conversion rate is orders divided by sessions. Revenue per $1 of ad spend is a dash where spend was $0.

Channel	Sessions	Orders	Conversion	Revenue	Ad spend	Revenue per $1 spent
Organic search	61,500	2,310	3.76%	$288,750	$4,000	$72.19
Paid search	38,200	1,420	3.72%	$163,300	$31,000	$5.27
Paid social	29,700	520	1.75%	$54,600	$22,500	$2.43
Email	21,400	1,640	7.66%	$213,200	$1,800	$118.44
Direct	33,200	1,010	3.04%	$119,180	$0	–

Table C · Revenue by month, last 12 months

Month	Revenue	Month	Revenue
Oct 2025	$612,400	Apr	$631,700
Nov	$735,900	May	$702,500
Dec	$904,200	Jun	$811,400
Jan 2026	$587,300	Jul	$868,900
Feb	$523,800	Aug	$797,600
Mar	$548,100	Sep	$839,030

Table D · Numbers the old dashboard led with

These are all real, but think about whether they help the decision.

Measure	Value
Total page views	612,000
Social media followers	48,300
Newsletter subscribers	31,900

© Yasas Sri Wickramasinghe

Team 4 · Rimu Outdoor Gear: sketch your layout

Pencil, five minutes, then photograph it

Laptop and projector (16:9)

Reader and decision, in one sentence

The most important item (top-left)

Notes: what moved, and why

© Yasas Sri Wickramasinghe

Design rationale

One row for every visual on your dashboard. This table is half of your three-minute pitch: it shows that each chart was chosen, not defaulted to.

READER AND DECISION

"Tom Nguyen needs to decide … looking at … , …"

#	Question it answers	Chart chosen	Chart rejected, and why	How colour is used
Ex.	Which classes at a gym are over-full or empty?	Horizontal bars of attendance as a share of capacity, sorted, with a 100% line	Pie chart: twelve thin slices, and a slice cannot show capacity	Grey bars. Accent only above 90% full
1				
2				
3				
4				
5				
6				
7				
8				

© Yasas Sri Wickramasinghe

TEAM 5 Tūī Valley Council · Brief

The public recycling dashboard

A local council serving 22,000 households · public sector

READER	SCREEN	HOW OFTEN
Residents of the Tūī Valley, most with no data skills, many reading on a phone	A phone in portrait, opened from a link on the rates letter	Once or twice a year, for about a minute

The council's goal is to divert 60% of household waste from landfill by 2030. Last year it collected 25,000 tonnes and 43.2% was kept out of landfill.

The council wants to publish its progress so residents can see how their suburb compares, and what a household can do about it. The audience is the whole community, so it must be understood without explanation, by someone who has never used a dashboard.

THE DECISION THIS DASHBOARD SUPPORTS

Do residents understand how the valley is doing, how their suburb compares, and the one thing they should change?

The reader must be able to answer, in five seconds

1. How much of our waste stays out of landfill, and how far is that from the 60% goal?
2. Are we improving year on year?
3. How does my suburb compare with the others?
4. What is the most common mistake, and what should I do instead?

What their dashboard gets wrong today

- The page is a 12-page PDF report with a donut chart split into 14 slices.
- Terms like "diversion rate", "contamination" and "tonnage" are never explained.
- Suburbs are listed alphabetically, so the best and worst are hidden in the middle.
- Light grey text on a white background is hard to read, particularly on a phone in sunlight.

Your design must

- Phone, portrait, about 390 × 844. One column. Thumb-friendly.
- Plain language. If a word needs explaining, replace it. No jargon on the screen.
- Meets accessibility basics: strong text contrast, and no meaning carried by colour alone.

STRETCH CHALLENGE

A resident should be able to find their own suburb in two taps. Show how you would highlight "you are here" without needing a search box or a long list.

© Yasas Sri Wickramasinghe

Your data

Copy these numbers into your chart tool. Work out any rates and totals you need from the counts. All figures are invented for this activity.

Table A · Waste collected last year, by stream (tonnes)

Everything except "Rubbish to landfill" is diverted from landfill.

Stream	Tonnes	Share of total
Rubbish to landfill	14,200	56.8%
Kerbside recycling	5,100	20.4%
Food scraps	2,400	9.6%
Green waste	3,300	13.2%

Table B · Share of waste diverted from landfill, by year

The 2030 goal is 60%.

Year	Diverted	Year	Diverted
2019	32.0%	2023	39.6%
2020	33.5%	2024	41.0%
2021	35.1%	2025	43.2%
2022	37.9%		

Table C · By suburb, last year

"Contaminated" means the share of recycling bins checked that held items that do not belong.

Suburb	Households	Diverted	Bins contaminated
Riverbend	3,200	52%	8%
Hillcrest	2,900	49%	11%
Old Town	1,800	47%	14%
Parkside	3,600	45%	12%
Lakeview	2,700	42%	17%
Stonefields	4,100	38%	19%
The Flats	2,300	34%	23%
Industrial Row	1,400	29%	27%

Table D · What goes wrong in contaminated recycling bins

Item	Share of contaminated bins
Soft plastics and plastic bags	28%
Food-soiled items such as pizza boxes	21%
Glass in the wrong bin	17%
Nappies and textiles	11%
Polystyrene	9%
Everything else	14%

© Yasas Sri Wickramasinghe

Team 5 · Tūī Valley Council: sketch your layout

Pencil, five minutes, then photograph it

Phone, portrait

Reader and decision, in one sentence

The most important item (top-left)

Notes: what moved, and why

© Yasas Sri Wickramasinghe

Design rationale

One row for every visual on your dashboard. This table is half of your three-minute pitch: it shows that each chart was chosen, not defaulted to.

READER AND DECISION

"Residents of the Tūī Valley needs to decide … looking at … , …"

#	Question it answers	Chart chosen	Chart rejected, and why	How colour is used
Ex.	Which classes at a gym are over-full or empty?	Horizontal bars of attendance as a share of capacity, sorted, with a 100% line	Pie chart: twelve thin slices, and a slice cannot show capacity	Grey bars. Accent only above 90% full
1				
2				
3				
4				
5				
6				
7				
8				

© Yasas Sri Wickramasinghe